

UTENA, Lithuania

WMO: 266330

Lat: 55.5153N

Lon: 25.5897E

Elev: 347

StdP: 14.51

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.5	1.8	-10.5	3.2	-4.5	-3.7	4.6	3.3	16.1	32.6	14.4	31.2	1.8	140	0.531

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	83.4	68.1	80.1	66.1	77.1	64.6	70.1	79.9	68.1	77.0	66.2	74.6	6.2	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.6	99.2	75.6	64.7	92.6	72.7	62.9	87.0	71.0	34.3	80.0	32.6	77.2	31.1	74.7	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.7	12.1	10.8	DB	-11.7	88.5	8.6	3.0	-17.9	90.7	-22.9	92.4	-27.8	94.1	-34.0	96.3	(4)
(5)			WB	-12.2	73.0	8.4	1.7	-18.2	74.2	-23.1	75.2	-27.8	76.1	-33.9	77.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.3	24.4	25.0	32.4	44.6	53.6	60.1	64.4	62.5	54.1	44.3	36.0	29.0	
	DBStd	16.21	11.14	11.18	8.29	7.83	7.73	6.17	5.54	5.57	6.17	7.05	8.06	9.67	
	HDD50	3596	794	700	546	200	49	2	0	0	30	204	420	651	
	HDD65	7693	1259	1120	1010	613	363	173	80	117	329	643	870	1116	
	CDD50	1515	0	0	1	37	160	304	446	387	154	26	0	0	
	CDD65	136	0	0	0	0	9	25	61	39	2	0	0	0	
	CDH74	1403	0	0	0	9	100	277	595	397	25	0	0	0	
	CDH80	297	0	0	0	0	11	47	151	87	1	0	0	0	
Wind	WSAvg	5.0	5.7	5.5	5.5	5.1	4.8	4.5	4.0	3.9	4.2	5.1	5.6	5.8	
Precipitation	PrecAvg	27.8	1.9	1.7	1.6	1.5	2.4	2.9	3.1	3.2	2.4	2.6	2.2	2.0	
	PrecMax	34.5	4.4	2.9	3.0	2.9	6.4	5.7	7.7	5.6	5.7	5.2	4.1	3.1	
	PrecMin	21.2	0.4	0.7	0.5	0.2	0.3	0.2	0.3	0.2	0.6	0.5	0.1	0.9	
	PrecStd	3.7	0.8	0.6	0.7	0.8	1.1	1.3	1.8	1.5	1.3	1.2	0.9	0.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.5	44.1	58.2	75.2	81.6	84.7	89.1	87.3	77.5	65.0	53.3	47.8	
		MCWB	43.1	40.5	47.0	59.3	65.0	67.3	71.2	70.4	63.4	57.0	49.7	45.6	
	2%	DB	41.3	41.3	51.8	69.5	76.8	80.8	84.3	81.9	72.3	60.5	50.5	44.9	
		MCWB	39.6	38.7	43.7	55.3	62.3	66.2	69.1	67.2	61.7	55.0	47.9	42.8	
	5%	DB	38.6	39.2	47.0	64.5	72.9	77.2	80.4	78.4	68.1	57.2	48.0	42.2	
		MCWB	37.0	37.2	41.6	52.0	59.7	64.3	67.2	65.1	59.5	53.2	45.9	40.6	
	10%	DB	36.6	37.3	43.4	59.6	68.9	73.5	76.8	74.7	64.9	54.6	46.0	39.7	
		MCWB	35.2	35.6	39.4	49.5	57.5	62.3	65.7	63.8	58.3	51.3	44.2	38.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.4	41.5	48.2	60.4	67.2	70.9	73.6	72.3	66.0	58.5	50.5	46.1
			MCDB	45.1	43.3	56.0	72.5	78.2	80.7	84.9	82.8	73.8	63.3	52.1	47.4
2%		WB	39.7	39.0	44.8	56.2	64.2	67.9	70.8	69.2	63.2	55.7	48.2	43.0	
		MCDB	41.2	40.7	49.7	67.5	74.2	77.2	80.9	78.9	69.8	59.4	50.2	44.4	
5%		WB	37.2	37.4	42.2	53.1	61.4	65.6	68.7	67.1	60.8	53.6	46.0	40.6	
		MCDB	38.4	39.0	46.2	61.8	70.5	74.9	77.5	75.4	66.4	56.6	47.8	42.1	
10%		WB	35.3	35.8	39.8	50.5	58.7	63.4	66.8	65.0	58.8	51.6	44.2	38.2	
		MCDB	36.5	37.3	43.1	58.2	66.9	71.5	74.9	72.6	63.8	54.5	45.9	39.6	
Mean Daily Temperature Range	5% DB	MDBR	7.9	9.9	12.9	18.6	20.1	18.5	18.4	19.0	16.4	11.4	6.9	6.4	
		MCDBR	6.8	7.6	18.9	27.5	25.3	24.6	24.7	25.2	22.3	14.7	8.7	6.8	
		MCWBR	6.2	6.3	12.4	15.1	12.6	12.2	12.0	12.2	13.2	10.2	7.3	6.4	
	5% WB	MCDBR	6.5	6.9	16.4	24.4	21.9	21.5	20.9	21.3	18.8	13.3	8.0	6.6	
		MCWBR	6.3	6.1	11.7	14.5	11.8	11.7	11.1	11.5	12.2	10.1	7.2	6.3	
Clear-Sky Solar Irradiance	taub	0.292	0.325	0.359	0.402	0.391	0.382	0.415	0.413	0.376	0.352	0.333	0.294		
	taud	2.189	2.166	2.201	2.211	2.313	2.367	2.294	2.293	2.370	2.400	2.269	2.212		
	Ebn at Noon	194	230	252	258	270	274	261	253	247	223	177	164		
	Edh at Noon	19	28	34	39	37	36	38	36	29	23	18	15		
All-Sky Solar Radiation	RadAvg	173	421	836	1282	1687	1755	1650	1361	927	448	156	106		
	RadStd	38	83	136	157	161	159	149	126	97	82	25	20		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.86	N/A	N/A	N/A	N/A	+1.22	N/A	-315	N/A	N/A
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page